

SALVAGE & OVERHAUL

Job Performance Requirements

NFPA 1001

4.3.14

5.3.4*

Note: All objectives in this section are testable.

Introduction

Subject: Salvage & Overhaul

Note:

1. Salvage refers to methods and operating procedures by which firefighters attempt to save property and reduce further damage from water, smoke, heat, and exposure, during or immediately after a fire. This may be accomplished by removing property from a fire area, by covering property, or by other means.
2. Overhaul refers to operations conducted once the main body of fire has been extinguished. These consists of: searching for and extinguishing hidden or remaining fire; placing building and its contents in a safe condition; determining the cause and origin of a fire; recognizing and preserving evidence of arson.
3. **Candidates will be required to notify:**
 - a. **“Proper Salvage techniques can begin at the time of initial fire attack or anytime during fire suppression.”**
 - b. **“Proper Overhaul techniques begin once the main body of fire has been extinguished.”**
4. **There are four (4) classifications of Fire Cause: Accidental; Incendiary; Natural; Unconfirmed.**
5. **Recognize evidence of Fire Cause and Arson such as: Multiple fires or points of origin; Trailers; Matches or lighters; Flammable or combustible liquids; Bottles or containers; Rubber or latex items; Candles or oily rags; Electrical sources or modified equipment; Fire patterns; Missing household property or personal items; Forced entry by anyone other than firefighters.**
6. **Potential or suspected evidence should be kept in place (unless situation or condition necessitates moving) and utilized in the condition found.**
Perimeter should be detected for evidence as well.
7. Techniques to preserve and protect evidence:
 - a. Follow department SOP's.
 - b. **Execute a perimeter and secure the scene.**
 - c. Restrict access to emergency personnel and investigators.
 - d. Isolate items.
 - e. **Prevent contamination (tainted) and spoliation (damaged or destroyed).**
8. Try to avoid contact or communicating with the property owner(s), renters, news media, or other bystanders. If necessary, an enough reply to any questions concerning fire cause shall be, **“The fire is under investigation.”**
9. **Detailed and precise listation is imperative.**
10. Several different control valves are utilized in fire suppression systems. Types of control valves include the following: OS&Y, PIV, WPIV, PIVA. Closing a valve to shut off the water supply is the preferred and safest method.

Performance Objective

Subject: Salvage & Overhaul

Task: Open a Ceiling and Check for Hidden Fire

Note: Shall be conducted under IDLH

Procedure:

1. Confirm order with OIC and ensure utilities are secured.
2. Select the appropriate tool(s).
3. Must convey proper actions.
4. Sound the floor for structural integrity prior to entering the building.
5. Demonstrate proper use of tool(s).
6. At the plane of entry, create a detection opening in the ceiling between the beams/trusses **(usually two feet apart)**. To achieve effectively, insert Pike Pole, rotate hook 90 degrees (hook shall be positioned at 3 or 9 o'clock), pull down and away.
7. While advancing in the building, keep Pike Pole pointed up and stand between area being pulled with back toward the means of egress (Escape Route).
8. Execute a second means of egress whenever possible.
9. Look for indicators of possible fire:
 - a. Charring
 - b. Cracks
 - c. Discolored and bubbled paint
 - d. Light fixtures and ceiling fans burned
 - e. Smoke
 - f. Sounds of fire burning wood
10. Locate the heaviest burned area or starting point of fire.
11. At every detection opening, search for concealed fire, burning ember or sparks, to ensure fire is demonstrated extinguished.
12. Pull enough ceiling down exposing burned area, until clear or unburned wood is visible.

Performance Objective

Subject: Salvage & Overhaul

Task: Open a Wall and Check for Hidden Fire

Note: Shall be conducted under IDLH

Procedure:

1. Confirm order with OIC and ensure utilities are secured.
2. Select the appropriate tool(s).
3. Must convey proper actions.
4. Sound the floor for structural integrity prior to entering the building.
5. Demonstrate proper use of tool(s).
6. Look for indicators of possible fire:
 - a. Charring
 - b. Cracks
 - c. Discolored and bubbled paint
 - d. Electrical outlets and light switches burned
 - e. Smoke
 - f. Sounds of fire burning wood
7. While advancing in the building, keep your back toward the means of egress (Escape Route).
8. Execute a second means of egress whenever possible.
9. **Create an detection opening in the wall between the studs (usually 16 inches apart). Create a second opening on the opposite side of fire stops (above & below).**
10. Locate the heaviest burned area or starting point of fire.
11. At every detection opening, search for concealed fire, burning ember or sparks, to ensure fire is demonstrates extinguished.
12. **Remove baseboards, casings, and framings, when necessary.**
13. Pull enough of the wall down exposing burned area, until clear or unburned wood is visible.

Performance Objective

Subject: Salvage & Overhaul

Task: Preserve and Protect Evidence of Fire Cause and Origin

Procedure:

1. Confirm order with OIC and convey proper actions.
2. **Recognize potential evidence and classification of fire cause:**
 - a. Accidental;
 - b. Incendiary;
 - c. Natural;
 - d. Unconfirmed.
3. **Preserve and Protect evidence:**
 - a. Avoid damaging, touching, disturbing, or tramping;
 - b. Avoid using excessive water during extinguishment when fire is under control;
 - c. **Leave evidence in place (unless situation or condition necessitates moving) and utilize in the condition found.**
4. **Execute a perimeter, secure the scene, and restrict access:**
 - a. Avoid contact with bystanders and reply to any questions with, **“The fire is under investigation”**;
 - b. Utilize security until an investigator arrives on scene.
5. **Prevent contamination and spoliation.**
6. **List all findings and actions regarding evidence:**
 - a. Record location, appearance, and time when found;
 - b. Record location, appearance, and time if moved;
 - c. Record actions involving events conducted at the scene, around and with evidence;
 - d. Record actions and information involving inability to preserve or protect;
 - e. Initiate chain of custody record and times.
7. **Provide evidence and records to investigator before leaving the scene.**

Performance Objective

Subject: Salvage & Overhaul

Task: Water Chute

Procedure:

1. Confirm order with OIC.
2. Selects two (2) Pike Poles.

Note: Can be setup without Pike Poles to use on stairwells.

3. Unfolds Salvage Cover demonstrates and positions the finished side facing down.
4. Place Pike Poles on the edge of both sides in the same direction, lying parallel to each other.
5. Ensure Pike Pole heads are beyond the top edge.
6. Wrap the Salvage Cover tightly around the Pike Poles while rolling toward the center.
7. Leave space between the Pike Poles for required width of application (Usually 2-3 feet).
8. Turn the Salvage Cover over, so the finished side is facing up.
9. Deploy the Water Chute by placing the hooks on a ladder rung, or another object, to secure in place.
10. Extend the Water Chute to desired location (Through a door, window, or in a catchall) for drainage.

Performance Objective

Subject: Salvage & Overhaul

Task: Water Catchall

Procedure:

1. Confirm order with OIC.
2. Unfolds Salvage Cover demonstrates and positions the finished side facing up.
3. Roll the sides of the longer edges inward approximately three (3) feet.
4. At each corner (4), fold over toward the center at a ninety (90) degree angle.
This will create an arrow shape at each end.
5. At each end, tightly roll the arrow until ends are resting on top of side rolls.
6. Tuck the end rolls under the side rolls to lock the corners.

Performance Objective

Subject: Salvage & Overhaul

Task: One Firefighter Salvage Cover Roll

Procedure:

1. Confirm order with OIC.
2. Unfolds Salvage Cover demonstrates and positions the finished side facing up.

Note: This objective is performed with two (2) firefighters simultaneously at opposite ends of the Salvage Cover.

3. Position yourself in the center at the end.
4. Grasp the cover midway between the center and the edge to be folded.
5. Using the other hand as a pivot, bring the midway point to the center.
6. Grasp the outside edge.
7. Using the other hand as a pivot, fold the outside edge over to the center and place it on top of the previous fold. Ensure all edges are in-line and dressed.

Note: If folded correctly, "Grommets" should be on top of folds.

8. On the other half of the Salvage Cover, repeat steps 4 thru 7.
9. Fold both ends over 12 inches.
10. Roll one end toward the other end to demonstrate.

Performance Objective

Subject: Salvage & Overhaul

Task: One Firefighter Salvage Cover Fold

Procedure:

1. Confirm order with OIC.
2. Unfolds Salvage Cover demonstrates and positions the finished side facing up.

Note: This objective is performed with two (2) firefighters simultaneously at opposite ends of the Salvage Cover.

3. Position yourself in the center at the end.
4. Grasp the cover midway between the center and the edge to be folded.
5. Using the other hand as a pivot, bring the midway point to the center.
6. Grasp the outside edge.
7. Using the other hand as a pivot, fold the outside edge over to the center and place it on top of the previous fold. Ensure all edges are in-line and dressed.

Note: If folded correctly, "Grommets" should be on top of folds.

8. On the other half of the Salvage Cover, repeat steps 4 thru 7.
9. With both firefighters at one end, grasp the corners and fold toward the center. Ensure to stop approximately 2 inches from center.
10. Continue to fold end in halves, placing it on top of the previous fold, until the width of the folds are approximately 12 inches. Ensure all folds are in-line and dressed.
11. On the other end of the Salvage Cover, repeat steps 9 & 10. You should have a gap of 4 inches between the two folded ends. The gap acts as a hinge.
12. Place one folded end on top of the other to demonstrate. The hinge allows the demonstrated fold to lie flat.

Performance Objective

Subject: Salvage & Overhaul

Task: Deploy a Salvage Cover Roll

Procedure:

1. Confirm order with OIC.
2. Start at one end of the object(s) to be covered.
3. **Unroll a sufficient amount to cover the end of the object(s).**
4. Lay the remaining roll on the end of the object(s) and continue to unroll toward the other end.
5. At one end grasp the top grommets, on the open edges, in each hand.
6. Snap the edges up and out, creating a balloon effect, to cover the object(s).
7. At the other end, use the same method.

Note: Steps 5 & 6 can be performed with two (2) firefighters simultaneously at opposite ends of the Salvage Cover.

8. Tuck in loose edges.

Performance Objective

Subject: Salvage & Overhaul

Task: Deploy a Salvage Cover Fold

Procedure:

1. Confirm order with OIC.
2. **Lay the fold on top and in the center of the object(s) to be covered.**
3. Unfold both sides to the opposite ends of the object(s).
4. At one end grasp the top grommets, on the open edges, in each hand.
5. Snap the edges up and out, creating a balloon effect, to cover the object(s).
6. At the other end, use the same method.

Note: Steps 4 & 5 can be performed with two (2) firefighters simultaneously at opposite ends of the Salvage Cover.

7. Tuck in loose edges.

Performance Objective

Subject: Salvage & Overhaul

Task: Stop Water Flow from a Sprinkler Head

Procedure:

1. Confirm with the Incident Commander (IC).
2. Verbalize:
 - a. Pendant – fanned larger teeth;
 - b. Upright – umbrella;
 - c. Sidewall – sideways or horizontal to wall.
3. If an individual sprinkler head is flowing water, utilize wedges or tongs by inserting into the sprinkler head between the arms.
4. Slide the wedges toward each other to stop the water flow.
5. Tongs must be opened by clamping the handles together to stop the water flow. Lock the tongs in the open position, with the keeper pulled toward the end of the handles. With Swivel-type tongs, turn the locking knob clockwise to lock in the open position.

Note: Water will still leak from the sprinkler head. Utilize one of the following methods to collect the leaking water beneath the sprinkler head:

- a. a bucket;
 - b. setup a catchall;
 - c. setup a water chute;
 - d. hang a fire hose directly under the sprinkler head and place the other end out of the building.
6. If the sprinkler system and/or multiple heads are flowing water, locate and close the control valve. It is imperative to ensure the fire is extinguished before closing the control valve.

Note: Bolt cutters may be needed to remove locks and chains on control valves.